USER REQUIREMENT DOCUMENT

WARDROBE AI

(AI Colour Match Dressing Assistant)



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Wardrobe AI

Version 1.0

Requirement Document

Version History

Version No	Date	Changed By	Changes Made
V1.0	31-08-2026	Karthikeya, Saval, Iliyaz, Sandeep	Initial Requirement Document
V1.1	02-09-2026	Karthikeya, Saval, Iliyaz, Sandeep	Added use cases and use case specifications
V1.2	04-09-2026	Karthikeya, Saval, Iliyaz, Sandeep	Updated functional and non-functional requirements

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Introduction

This document describes the requirements of the **Wardrobe AI – AI Colour Match Dressing Assistant**.

Wardrobe AI is an AI-based fashion assistance system that helps users choose suitable clothing combinations. The system allows users to upload images of their clothes and uses computer vision and colour analysis techniques to identify clothing types and colours.

The system provides colour-matching recommendations and evaluates combinations such as shirts with pants, dresses with accessories, and complete outfits.

Wardrobe AI is designed to help users make better dressing decisions and improve their confidence when selecting an outfit for different occasions.

Purpose

The purpose of this document is to gather and define the requirements needed to implement the Wardrobe AI system.

The system aims to:

- Identify clothing items from uploaded images.
- Detect the dominant and secondary colours of clothing.
- Recommend suitable colour combinations.
- Evaluate combinations of clothing items.
- Provide an outfit match score.
- Suggest alternative clothing combinations.
- Provide recommendations based on occasions.
- Help users make dressing decisions without depending on another person.

The main purpose of Wardrobe AI is to provide a simple and intelligent **personal dressing assistant** using Artificial Intelligence and Computer Vision.

Intended Audience

The intended audience of Wardrobe AI includes:

- Students
- Working professionals
- Job seekers
- Fashion-conscious users
- People who have difficulty selecting matching clothes
- Users preparing for interviews, parties, weddings and other events
- Anyone who wants quick assistance while selecting an outfit

The system is intended for typical users who are familiar with smartphones, computers and basic image uploading.

Stakeholders

Client

Wardrobe AI Project Team

Users

Individuals who use Wardrobe AI to analyse and select their outfits.

Developers

The development team responsible for implementing the AI, backend and frontend components.

Project Guide / Faculty

Person responsible for reviewing and evaluating the project.

Product Vision

Vision Statement

The vision of Wardrobe AI is to develop a **user-friendly AI-powered personal dressing assistant** that helps users select suitable clothing combinations with confidence.

Wardrobe AI analyses the user's clothing images, identifies clothing categories and colours, evaluates colour compatibility and provides easy-to-understand recommendations.

The system should reduce the uncertainty users experience when deciding **“Does this outfit look good together?”**

Technologies

The proposed technologies for the Wardrobe AI system are:

Frontend

- HTML
- CSS
- JavaScript
- React
- Tailwind CSS

Backend

- Python
- Flask / FastAPI

Artificial Intelligence

- YOLO
- OpenCV
- NumPy
- Pillow
- Scikit-learn
- Computer Vision
- Colour analysis algorithms

Optional AI

- Gemini API

- CLIP / multimodal models

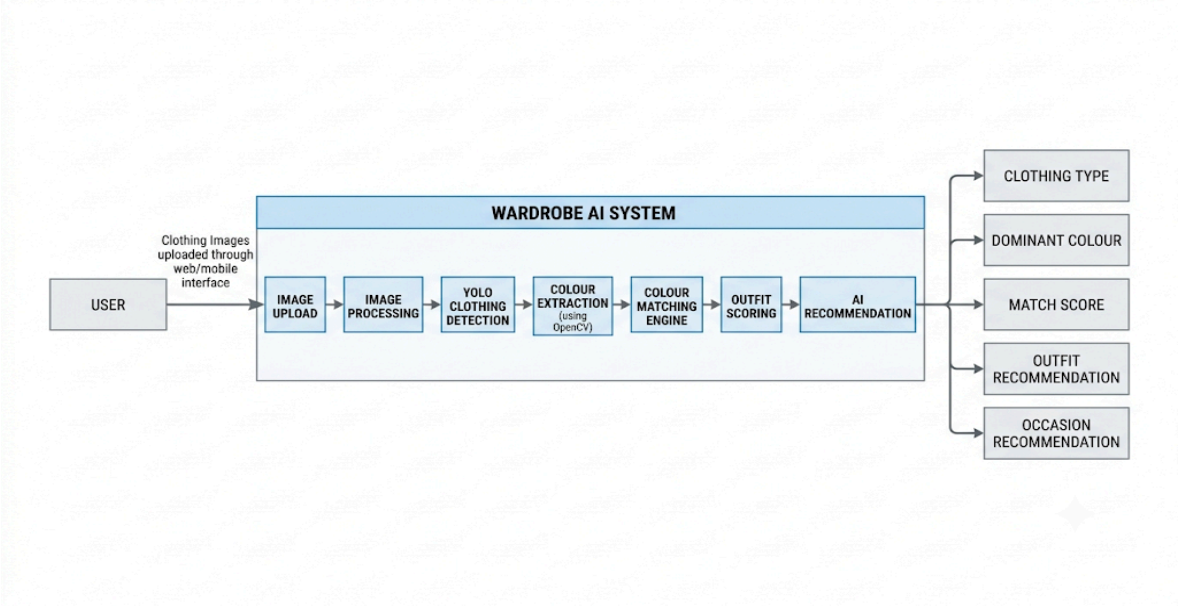
Database

- MongoDB Atlas

Development Tools

- Visual Studio Code
- Python
- Git / GitHub

System in Context



Wardrobe AI is an intelligent multi-user system that assists users in selecting suitable clothing combinations.

The user uploads an image of a clothing item. The system processes the image using computer vision techniques.

The system performs the following major operations:

1. Receives clothing image from the user.
2. Detects the clothing item.
3. Identifies the clothing category.
4. Extracts the dominant colour.
5. Identifies secondary colours when applicable.
6. Compares clothing colours.
7. Generates matching recommendations.

8. Calculates an outfit match score.
9. Provides suggestions for improving the outfit.
10. Provides occasion-based recommendations where applicable.

Example

User uploads:

Blue Shirt + Black Pant

The system analyses the two items and may return:

Match Score: 95/100
Excellent Combination
Black provides a strong neutral match with the blue shirt.

The system may also suggest:

- White sneakers
- Brown or black belt
- Silver accessories

User Characteristics

The users of Wardrobe AI are expected to be typical smartphone or computer users.

Users should have:

- Basic knowledge of smartphones/computers.
- Ability to upload images.
- Basic understanding of clothing.
- Ability to select clothing images.
- Ability to read simple recommendations.

No advanced technical knowledge is required.

Constraints

The following constraints may affect the system:

1. The accuracy of clothing detection depends on image quality.
2. Poor lighting may affect colour detection.
3. Very complex backgrounds may affect clothing detection.
4. Similar colours may be difficult to distinguish.
5. The recommendation system may not represent every individual's personal fashion preference.
6. Internet connectivity may be required for cloud-based AI features.
7. AI recommendations are suggestions and should not be considered absolute fashion rules.
8. System performance depends on available computing resources.

System-Wide Requirements

Actors

The system primarily interacts with the following actors:

1. User

The user uploads clothing images and receives colour matching and outfit recommendations.

2. AI System

The AI system analyses clothing images, detects clothing types and colours, calculates compatibility and generates recommendations.

3. Administrator

The administrator may manage system information, monitor the system and maintain application data.

Events

The most important events in Wardrobe AI are:

1. User opens Wardrobe AI.
2. User uploads a clothing image.
3. System detects clothing type.
4. System identifies clothing colour.
5. User selects another clothing item.
6. System compares clothing colours.
7. System generates a match score.
8. System provides matching recommendations.
9. User requests a complete outfit recommendation.
10. User selects an occasion.
11. System provides occasion-based recommendations.
12. User views the final outfit recommendation.

User Visible Events

No.	Actor	Action	Object	Frequency	Arrival Pattern	Response
1	User	Opens	Wardrobe AI	Daily	Episodic	Home page is displayed
2	User	Selects	Upload Clothing	Daily	Episodic	Image upload interface is displayed
3	User	Uploads	Clothing Image	Daily	Episodic	Image is sent for analysis
4	User	Selects	Clothing Type	Daily	Episodic	System processes selected clothing
5	AI System	Detects	Clothing Item	Per request	Episodic	Clothing category is identified
6	AI System	Extracts	Colour	Per request	Episodic	Dominant colour is identified
7	User	Uploads	Second Clothing Item	Daily	Episodic	System receives second item
8	User	Selects	Match Outfit	Daily	Episodic	System compares clothing items

9	AI System	Calculates	Match Score	Per request	Episodic	Compatibility score is generated
10	AI System	Provides	Recommendation	Per request	Episodic	Suitable matching colours are displayed
11	User	Selects	Occasion	Occasionally	Episodic	Occasion is selected
12	AI System	Generates	Outfit Recommendation	Per request	Episodic	Suitable outfit is displayed
13	User	Views	Result	Per request	Episodic	User sees final recommendation

Functional Requirements

The Wardrobe AI system shall provide the following functional requirements:

FR-01: Upload Clothing Image

The system shall allow users to upload an image of a clothing item.

FR-02: Clothing Detection

The system shall detect clothing items present in the uploaded image.

FR-03: Clothing Classification

The system shall identify clothing categories such as:

- Shirt
- T-shirt
- Pant
- Dress
- Shoes
- Jacket
- Other supported clothing categories

FR-04: Colour Detection

The system shall identify the dominant colour of the detected clothing item.

FR-05: Secondary Colour Detection

Where applicable, the system shall identify secondary colours present in the clothing.

FR-06: Colour Matching

The system shall compare the colours of two or more clothing items.

FR-07: Match Score

The system shall generate a numerical compatibility score for a clothing combination.

Example:

Match Score: 92/100

FR-08: Recommendation

The system shall recommend suitable colours and clothing combinations.

FR-09: Outfit Evaluation

The system shall evaluate a complete outfit containing multiple clothing items.

FR-10: Improvement Suggestions

The system shall provide suggestions when the selected combination has low compatibility.

Example:

> Try replacing the brown pants with black or beige pants.

FR-11: Occasion Recommendation

The system shall provide recommendations based on occasions such as:

- Interview
- Office
- Party
- Wedding
- Casual outing
- College

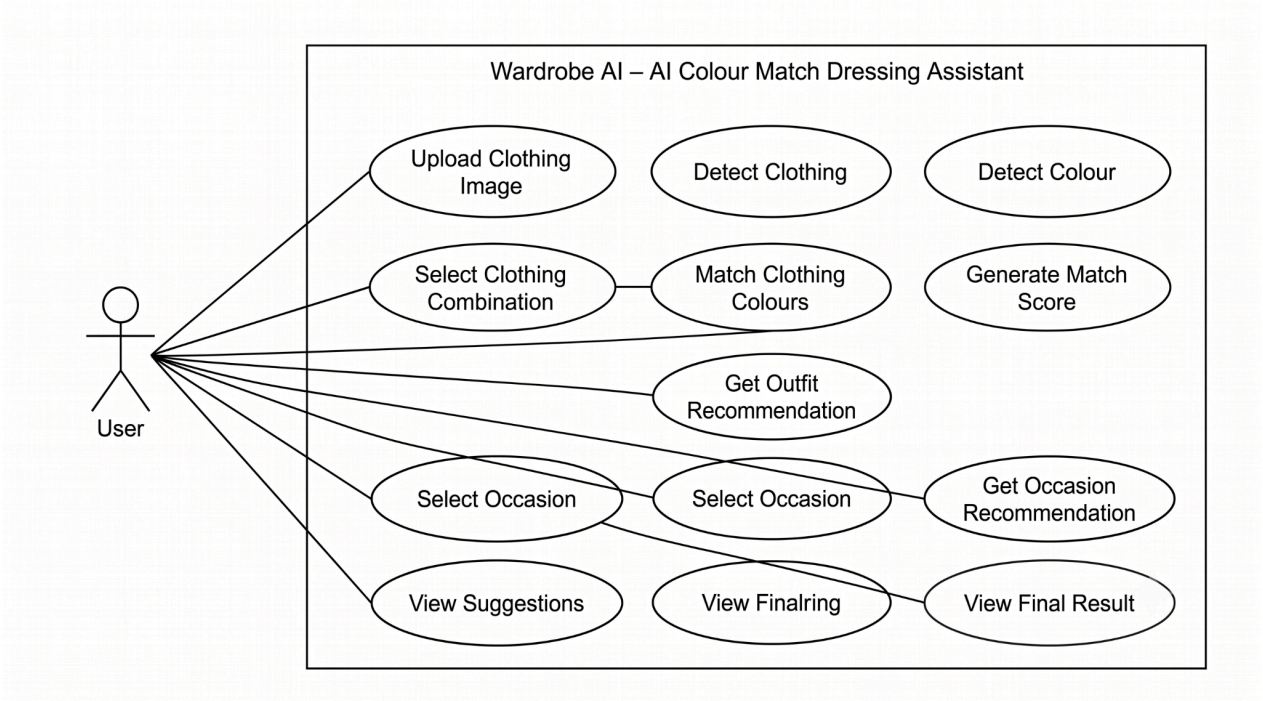
FR-12: Result Display

The system shall display the detected clothing, colours, match score and recommendations in an understandable format.

Use Case Overview

No.	Use Case ID	Use Case Name	Priority	Stability	Verifiability
1	UC-WAI-UCI	Upload Clothing Image	High	Stable	Verifiable
2	UC-WAI-DCT	Detect Clothing	High	Stable	Verifiable
3	UC-WAI-CDC	Detect Clothing Colour	High	Stable	Verifiable
4	UC-WAI-CM	Match Clothing Colours	High	Stable	Verifiable
5	UC-WAI-MS	Generate Match Score	High	Stable	Verifiable
6	UC-WAI-OR	Outfit Recommendation	High	Stable	Verifiable
7	UC-WAI-OCR	Occasion Recommendation	Medium	Stable	Verifiable
8	UC-WAI-OS	Outfit Scoring	High	Stable	Verifiable
9	UC-WAI-AS	Alternative Suggestions	Medium	Stable	Verifiable
10	UC-WAI-VR	View Recommendation	High	Stable	Verifiable

Use Case Diagram



Your main use-case diagram should show:

User connected to:

- Upload Clothing Image
- Detect Clothing
- Detect Colour
- Select Clothing Combination
- Check Match
- View Match Score
- Get Outfit Recommendation
- Select Occasion
- View Suggestions

And the **AI System** connected internally to:

- Image Processing
- Clothing Detection
- Colour Extraction
- Colour Matching
- Outfit Scoring
- Recommendation Generation

Use Case Specifications

1. UC-WAI-UCI: Upload Clothing Image

Actor: User

Description:
Allows the user to upload an image of a clothing item for analysis.

Main Flow

1. User opens Wardrobe AI.
2. User selects Upload Clothing.
3. System opens image selection interface.
4. User selects an image.
5. System validates the image.
6. System uploads the image.
7. System begins image analysis.

Alternative Flow

If the uploaded file is unsupported, the system displays an error message and asks the user to upload another image.

Output

Uploaded clothing image is successfully received by the system.

2. UC-WAI-DCT: Detect Clothing

Actor: AI System

Description:
The system identifies clothing items present in an uploaded image.

Main Flow

1. System receives image.
2. Image is processed.
3. AI model detects clothing objects.
4. Bounding/segmentation information is generated.
5. Clothing category is identified.
6. Result is returned to the user.

Output

Example:

Detected Item: Shirt

3. UC-WAI-CDC: Detect Clothing Colour

Actor: AI System

Description:
The system analyses the detected clothing region and identifies its dominant colour.

Main Flow

1. System receives detected clothing region.
2. Background is reduced or ignored.
3. Colour information is extracted.
4. RGB/HSV values are analysed.
5. Dominant colour is calculated.
6. Colour name is generated.

Output

Example: Clothing: Shirt —- Dominant Colour: Blue

4. UC-WAI-CM: Match Clothing Colours

Actor: User / AI System

Description:
The system compares two or more clothing items and determines whether their colours work well together.

Main Flow

1. User uploads first clothing item.
2. System identifies its colour.
3. User uploads second clothing item.
4. System identifies its colour.
5. Colour compatibility is calculated.
6. System generates a matching result.
7. Result is displayed.

Output

Blue Shirt + Black Pant
Excellent Combination

5. UC-WAI-MS: Generate Match Score

Actor: AI System

Description:
The system calculates a numerical score representing the compatibility between selected clothing items.

Main Flow

1. System receives clothing colours.
2. Colour relationships are analysed.
3. Compatibility rules/model are applied.
4. Score is calculated.
5. Score is displayed.

6. UC-WAI-OR: Outfit Recommendation

Actor: User / AI System

Description:
Provides recommendations for combining multiple clothing items.

Main Flow

1. User uploads clothing items.
2. System identifies each clothing item.
3. System identifies colours.
4. System evaluates compatibility.
5. System generates outfit recommendation.
6. User views the recommendation.

Example

Blue Shirt
Black Pant
White Sneakers

Overall Rating: 9.3/10

7. UC-WAI-OCR: Occasion Recommendation

Actor: User

Description:
Allows the user to select an occasion and receive appropriate dressing recommendations.

Supported Occasions

- Interview
- Office
- Wedding
- Party
- Casual
- College

Main Flow

1. User selects an occasion.
2. User provides clothing items.

3. System analyses the clothing.
4. System considers the selected occasion.
5. System generates recommendations.
6. Recommendation is displayed.

8. UC-WAI-OS: Outfit Scoring

Actor: AI System

Description:
The system evaluates a complete outfit and generates an overall score.

Main Flow

1. User selects multiple clothing items.
2. System analyses each item.
3. Colour compatibility is calculated.
4. Clothing combination is evaluated.
5. Overall score is generated.
6. Result is displayed.

Example

Overall Rating: 9.3/10 – Great outfit combination.

9. UC-WAI-AS: Alternative Suggestions

Actor: AI System

Description:
The system provides alternative clothing suggestions when the selected combination is not suitable.

Example

Selected:

Blue Shirt + Brown Pant

System:

Match Score: 45/100

Suggestion:

Try black, beige or grey pants instead.

10. UC-WAI-VR: View Recommendation

Actor: User

Description:
Allows the user to view the final AI-generated dressing recommendation.

The result may contain:

- Clothing type
- Dominant colour
- Match score
- Overall outfit rating
- Recommended colours
- Alternative suggestions
- Occasion suitability

Non-Functional Requirements

Reliability

The system should provide consistent results for similar input images.

The system should handle invalid images without crashing.

If an AI model fails to detect a clothing item, the system should provide an appropriate message to the user.

Usability

Wardrobe AI should have a simple and user-friendly interface.

The user should be able to upload clothing images with minimum steps.

Results should be presented clearly using:

- Clothing names
- Colour names
- Scores
- Recommendations
- Simple explanations

The system should be understandable even for users with limited technical knowledge.

Availability

The system should be available whenever the application server is running.

The system should provide appropriate error messages when AI services or external services are temporarily unavailable.

Accessibility

The interface should use readable text and clearly identifiable buttons.

Important information such as match scores and recommendations should be visually distinguishable.

The system should be usable on both desktop and mobile-sized screens.

Performance

The system should process uploaded images within a reasonable amount of time.

The AI model should provide clothing detection and colour analysis without unnecessary delay.

The frontend should provide feedback while image processing is taking place.

For example:

> **Analysing your outfit...**

Security

The system should protect user-uploaded images.

Uploaded images should not be exposed to unauthorized users.

User information should be securely stored if user accounts or history are implemented.

The system should validate uploaded files to prevent unsupported or malicious file uploads.

Project Scope

Phase 1 – Basic Clothing Analysis

- Upload clothing image
- Detect clothing
- Identify clothing type
- Detect dominant colour

Phase 2 – Colour Matching

- Upload shirt
- Upload pant
- Compare colours
- Generate match score
- Recommend matching colours

Phase 3 – Complete Outfit

- Shirt
- Pant
- Shoes
- Accessories
- Overall outfit score
- Improvement suggestions

Phase 4 – Smart Recommendations

- Occasion-based recommendations
- More intelligent AI recommendations
- Personalised suggestions

Platform Compatibility

Wardrobe AI should support modern web browsers including:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

The application should be designed to work on:

- Desktop
- Laptop
- Tablet

- Mobile devices

Expected Output

The final Wardrobe AI system should allow a user to upload clothing images and receive a result similar to:

Detected Clothing
👕 Shirt – Blue
👖 Pant – Black

Match Score
⭐ 95/100

Result
Excellent Combination

Suggestions
👟 White Sneakers
⌚ Silver Watch
👔 Brown/Black Belt

The overall objective is to make the user feel more **confident and comfortable about their dressing choices**, especially when attending interviews, offices, parties, weddings and other events.